

# PAPER\_526 — 3D-IPO Non-Linear Three-Helix Progression Overlay

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**CP4 Class:** ThreeDIPONonLinearProgressionCalculator (#121)

**Quality Score (QS):** 5 / 5

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## Abstract

This paper presents a UQFF analysis of 3D-IPO Non-Linear Three-Helix Progression Overlay, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## §1 — Overview

The **3D-IPO** (Three-Dimensional Irrational-Progress Overlay) framework models the simultaneous progression of three distinct physical axes as interlocking helices in a 26-dimensional UQFF braid space:

| Helix | Axis                | Description                                           |
|-------|---------------------|-------------------------------------------------------|
| $H_1$ | Wolfram progression | Computational irreducibility path through state space |
| $H_2$ | $\pi$ progression   | Irrational decimal expansion trajectory               |
| $H_3$ | F_U_Bi_i axis       | Buoyancy-force magnitude sequence                     |

Because  $\pi$  is irrational, the crossing pattern of  $H_1$  and  $H_2$  never repeats — generating a **topologically unique braid fingerprint** for every physical system.

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## §2 — Core Equation

$$n_{\text{cross}} = \arg \min_n |W_{\text{prog}}(n) - \Pi_{\text{prog}}(n) \cdot F_{U\_Bi}(x)|$$

where:

| Symbol                 | Definition                                                |
|------------------------|-----------------------------------------------------------|
| $W_{\text{prog}}(n)$   | Wolfram computation depth at step $n$                     |
| $\Pi_{\text{prog}}(n)$ | $\lfloor 10^n \pi \rfloor \bmod 10$ — $n$ -th $\pi$ digit |
| $F_{U\_Bi}(x)$         | UQFF buoyancy force at parameter $x$                      |
| $n_{\text{cross}}$     | First crossing index (braid primary node)                 |

## §3 — UQFF Number Systems Integration (PAPER\_429)

### Vacuum Density Series (VDS)

$$A_{\text{helix}} = \text{Li}_{26}([SSq]) = \sum_{k=1}^{26} \frac{[SSq]^k}{k^{26}} \approx 0.570$$

VDS normalises each helix amplitude, ensuring all three axes share the same 26-dimensional natural units. The common amplitude anchor prevents artificial scale separation between Wolfram,  $\pi$ , and F\_U\_Bi\_i progressions.

### Dipole Vortex Primes (DVP)

$$\Delta_{\text{vortex}} = p_{\text{special}} = 113 \quad (p > 26)$$

Prime 113 governs the vortex node spacing on all three helix axes. As the first prime beyond the 26-dimensional scale of UQFF, it encodes the shortest non-reducible interval between physically distinct crossing events.

## §4 — Braid Topology Proof

**Proposition:** The 3D-IPO braid has no repeating sub-word.

*Proof sketch:*

1.  $H_2$  is driven by  $\pi$  digit sequence, which is conjectured (and computationally verified to 100 trillion digits) to be **normal** — every finite digit string appears with equal frequency but never periodically.
2.  $H_1$  follows Wolfram computation depth, which by the **Principle of Computational Irreducibility** cannot be compressed to a shorter rule.
3. The crossing condition  $W_{\text{prog}}(n) = \Pi_{\text{prog}}(n) \cdot F_{U\_Bi}(x)$  requires simultaneous coincidence in two irreducible sequences  $\rightarrow$  probability of periodicity = 0.

$$P(\text{braid repeats}) = 0$$

## §5 — Available Equations

| Equation                                               | Description                      |
|--------------------------------------------------------|----------------------------------|
| $\text{braid}(n) = \sum_{\text{helix}} e^{i\pi n/p_k}$ | Phase-space braid representation |
| $\rho_{\text{cross}} \propto 1/\log(n)$                | Non-repeating crossing density   |
| $A_k = \text{Li}_{26}([SSq])$                          | Helix amplitude from VDS         |
| $\Delta_k = p_{\text{special}} = 113$                  | Vortex node spacing from DVP     |

## §6 — Observational Implications

- **Galaxy rotation curves:** Each galaxy leaves a unique 3D-IPO fingerprint in its UQFF buoyancy field, distinguishing it from all other systems.
- **Pulsar timing:** Pulsar spin-down sequences correspond to  $H_3$  axis samples; non-repeating braid predicts no exact period doubling.

- **Wolfram Hypergraph:** 3D-IPO crossing events correspond to causal cone intersections in the Wolfram physics hypergraph (SOURCE116).

## §7 — CP4 Calculator Output

```
calc = ThreeDIPONonLinearProgressionCalculator()
result = calc.compute(dataset={'SSq': 0.57}, n_steps=1000)
# result['n_cross'] - first crossing index
# result['crossing_count'] - total crossings in n_steps
# result['braid_topology'] - 'NON_REPEATING (irrational π)'
```

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.121 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 59, \quad n_{\text{channel}} = 7/26$$

Since  $p_{\text{DVP}} = 59$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is  **$10^4 \text{ yr}$**  (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework  | Canonical Value                                | This Paper                             | Status                 |
|------------|------------------------------------------------|----------------------------------------|------------------------|
| VDS ratio  | $\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.121                | ✓ Threshold-consistent |
| DVP prime  | $p_k \in \{2,3,\dots,113\}$                    | $p_{\text{DVP}} = 59$                  | ✓ Resonant             |
| BSH layers | 26 harmonic terms                              | $j = 1 \dots 26,$<br>$\cos(2\pi j/26)$ | ✓ Full 26D projection  |

| Framework      | Canonical Value                       | This Paper                 | Status      |
|----------------|---------------------------------------|----------------------------|-------------|
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Applied in VDS exponential | ✓ Canonical |
| [SSq]          | 0.57                                  | Applied in BSH saturation  | ✓ Canonical |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                   | UQFF Prediction                                                                     | SM / Experiment                                  | Source                                          | Alignment                                                    |
|------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| Higgs mass<br>$m_H$          | UQFF<br>$K_{\text{HIGGS}}=47.34 \rightarrow$<br>$m_H \text{ UQFF} =$<br>125.09 GeV  | $m_H = 125.20 \pm$<br>0.11 GeV                   | PDG<br>2024                                     | 99.8%                                                        |
| Cosmological<br>$\Lambda$    | UQFF                                                                                | $\nabla \text{UA}$                               | $^2 \rightarrow$<br>1.09e-52<br>$\text{m}^{-2}$ | $\Lambda =$<br>1.114e-52<br>$\text{m}^{-2}$<br>(Planck+DESI) |
| Thomson $\sigma_T$<br>(QED)  | UQFF $U_m$ kernel:<br>$\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$                | $\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$   | PDG<br>2024                                     | 100% (exact)                                                 |
| $\kappa$ baryon<br>stability | $\kappa = 0.0005/\text{day};$<br>scale separation<br>$10^{33}$ from proton<br>decay | $\tau_p > 7.7\text{e}33 \text{ yr}$<br>(Super-K) | Super-K<br>2024                                 | ✓ UQFF<br>baryon-safe                                        |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200$  PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

*Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## §8 — References

- PAPER\_429: Three New UQFF Number Systems (VDS / DVP / BH)
- SOURCE116: Wolfram Hypergraph Emergent Spacetime
- grok\_share\_2515709ed.txt: BigBangHypergraphTheory Millennium proof set
- Wolfram, S. (2020): *A Project to Find the Fundamental Theory of Physics*